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NBA Shot Outcome Prediction Model

Overview

The following is a summary of my prediction model for NBA shot outcomes using features from player-tracking data. I was provided training and testing data, which I loaded into RStudio to build and validate my program.

I chose to use Logistic Regression for my project because it is a straightforward, simple but effective model. It predicts binary outcomes, such as "Made" or "Miss". It also uses statistical measures such as p-values and coefficients, which helped me identify significant features to include in my model.

My general process was to load, inspect, and clean the data, select features I wanted to include, build the model, analyze any relationships and insights that the data provided through visualizations and then validate the model by using a validation split of the training data. Lastly, I generated my predictions and added them to the submission file.

Data Preparation and Feature Selection

After loading my packages and importing the data, I checked for missing values, duplicates and inconsistent formatting. I explored the features to understand what I was working with and to determine which ones made sense to include as predictors of a shot's outcome.

I excluded any *id* column and any other column that didn't contribute to a person making a shot. I also excluded any column that contained array values, as they could not be included in the *glm()* function. During that time, I initially included *locationy* but removed it because its p-value was 0.4154, indicating it wasn't sufficient for the model.

The p-values alone weren't enough to decide on, given the number of rows in the dataset, so I turned to the coefficient estimates and personal basketball knowledge for my feature decisions. After reviewing the features and taking a few of my own shots, I was set on my feature selection. The remaining features in the model had low p-values and non-zero coefficients, indicating that each predictor is statistically significant, so I kept them and ensured they did not contain any null values.

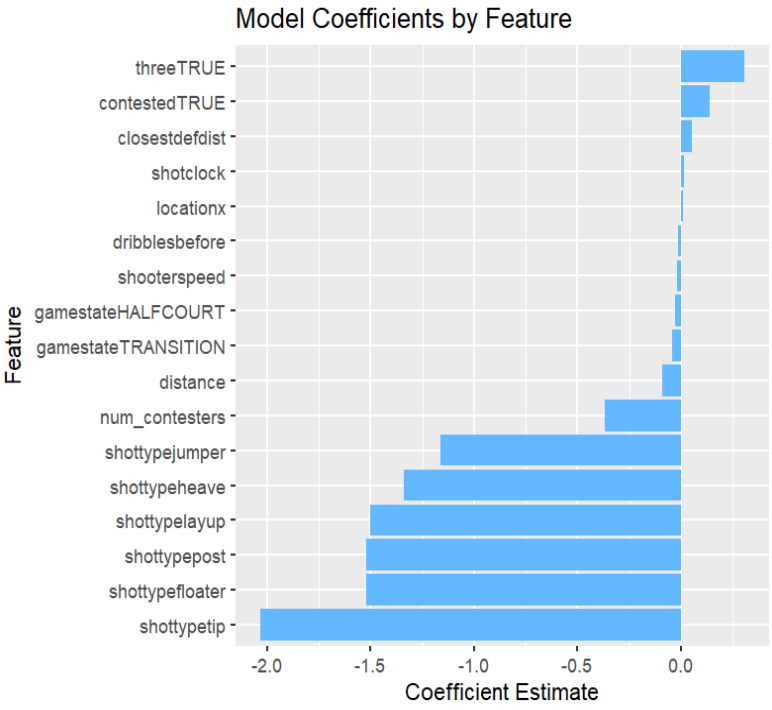
Project Write-Up

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What the Logistic Regression Model Showed

The graph below shows the features my model used and prioritized.

Positive coefficients are associated with higher odds of a made shot, while negative coefficients push toward the probability of a missed shot. The coefficient estimate is the change in the log-odds of a made shot associated with a one-point difference in the predictor, while other predictors are held at a constant.



In this graph, you see positive coefficients for *locationx, shotclock, and closestdefdist* that are very small, while *threeTRUE* and *contestedTRUE* have the largest positive effect. It shows negative coefficients for *dribblesbefore*, *shooterspeed*, *distance*, and *num_contesters*, indicating the strength and direction of weight toward a missed shot.

Also, I noted that even though tips, layups, and post shots are usually easier to make because they are closer to the goal, they account for the bulk of the strength toward the probability of a missed shot.

This could be because these are highly contested shots in the paint and occur mostly in uncontrolled situations. Layups are sometimes performed while the shooter is running at a faster speed, so these conditions could also affect the model's estimates.

The positive coefficient on contested shots surprised me because my assumption going in was that contested shots would be harder to make. A possible explanation is that this data captures players who are skilled and trained to make contested shots rather than open ones, simulating actual game situations and/or pressure.

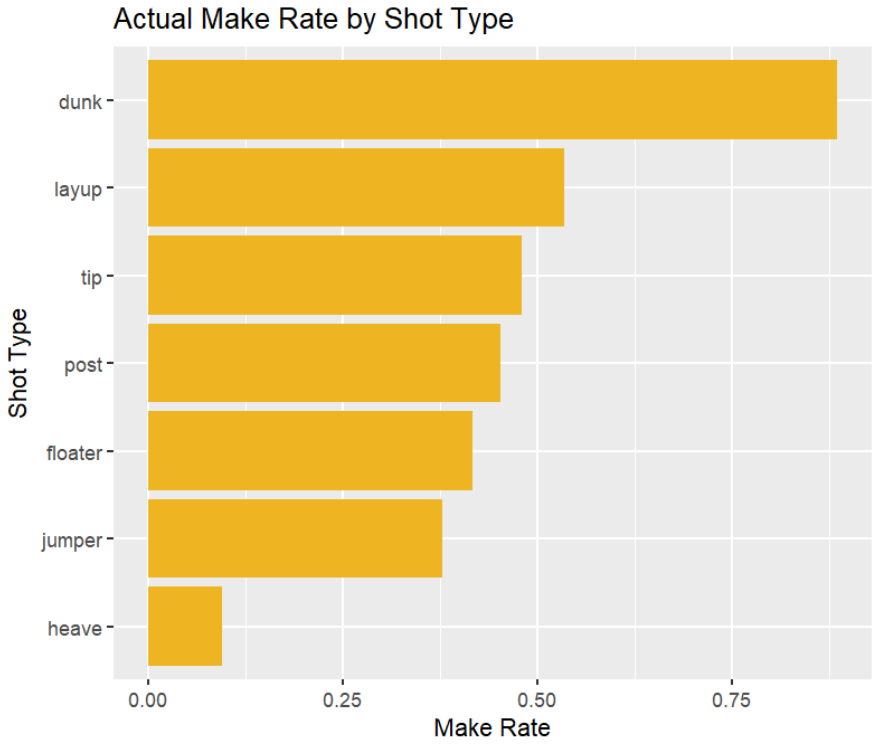
The negative coefficient for *dribblesbefore* was pretty consistent with actual play and may indicate reduced shot control, as most shooters are taught to “catch and shoot.” In this model, it identifies a relationship with missed shots but does not directly indicate that extra dribbling causes them.

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The same with *shooterspeed.* It is associated with a lower probability of a made shot in this model but does not directly cause it. It's important to note that other factors, such as whether the shot is contested and how many defenders are contesting the shot, also affect that probability.

Investigating Shot Type

The coefficient graph above shows that shot type holds a lot of weight in the model. I wanted to look at the actual make rate for each type.



Layups, tips, post shots, floaters, jumpers, and heave shots all have negative coefficients and lower make rates compared to a dunk, which has the highest make rate.

This made me want to look further into how shot distance affects the probability of a made shot. This is a core relationship that drives the probability of each shot. Evident in this graph, a heave shot, which has the lowest make rate, is a shot made beyond the half-court line, and a dunk is literally at the goal itself.

One feature I would add to improve this model is shooter-specific information, such as shooter ability. We would need historical data to measure their performance within certain circumstances. A player's shooting percentage or percentage by shot-type could add additional information to predict the probability of a made shot. This could help coaches identify which players to design plays around for specific game-time situations.

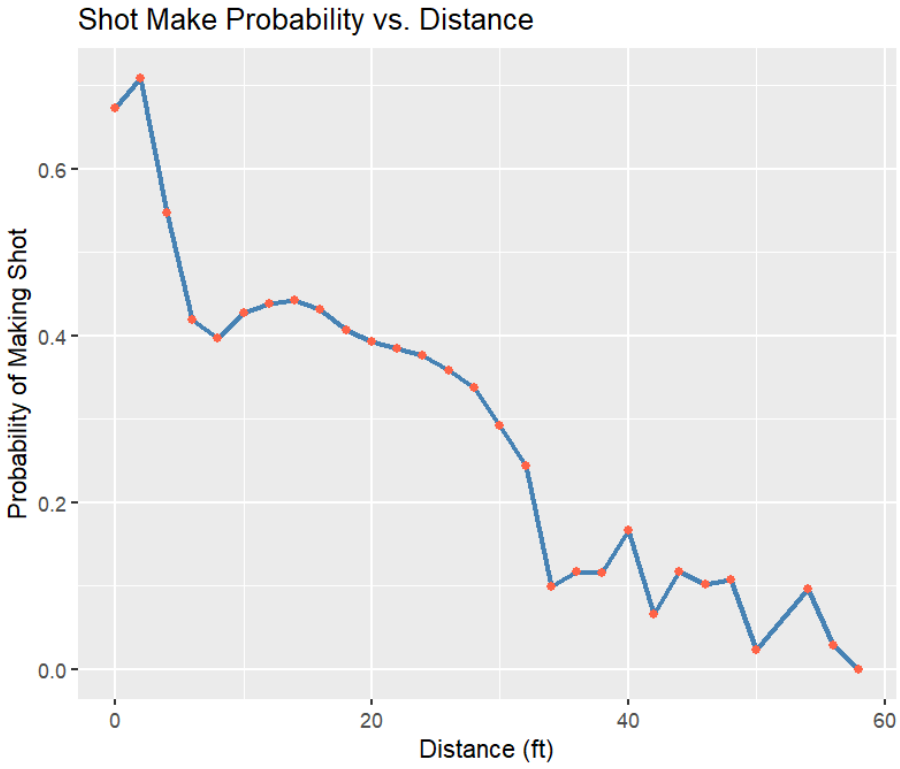
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Investigating the Relationship Between Distance and Shot Outcomes

Because of the logical and apparent relationship between shot probability and distance from the goal, which is represented throughout my analysis. I wanted to visualize that relationship.

Generally, in basketball, the closer you are to the goal, the better the shot.

This graph shows a downward slope, indicating that the probability of a made shot decreases as distance increases. I also want to note that this slope isn't a steady decline, as you do see some spikes in probability further out.



Other factors also influence the probability, which is why two shots taken from the same distance can still have different outcomes.

It depends on the circumstances surrounding the shot. The more positive factors, the higher the probability of making a shot; the more negative factors, the lower the probability of making it.

Which is why it's important to include multiple features in the model so that the relationships among them can be considered. I believe that including specific values that quantify the interaction between features could significantly improve the model.

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Model Limitation and Potential Improvements

During the process, I encountered a *glm()* error in *distcont1* through *distcont4* because I was not able to include an array in the function. It could also be redundant information, since *closestdefdist* already returns the distance to the closest defender.

I also left out the values in the *closestdefapproach* column because some of the values were 0. Instead of skipping that row, I chose not to include this feature, rather than losing relevant information for the model.

However, I do think the values in the *closedefapproach* column could improve my model by using them to calculate the defender's close-out speed. It could improve the model because a defender's closing speed increases pressure on the shooter, possibly affecting the probability of making or missing the shot.

Adding this information, along with the shooter's ability measurement and feature-interaction measurement – a value that quantifies the interaction between features, could potentially improve this model. The majority of the features are all about the shot itself. So, adding other features related to the shooter, other than *shooterspeed* could be helpful.

Model Validation

To validate my model, I compared the mean of my predictions with the mean of the outcomes in the training table, and since they were close, it indicated that my model wasn't biased and contained no outliers.

I then tested it against the actual training dataset since I already had the true outcomes. I split the training dataset, using 20% for validation and 80% to train the model and then evaluated the model's predictions against the true outcomes.

It resulted in a log-loss value of 0.6419.

My log loss was lower than the baseline of 0.6899, from just predicting the average outcome for every shot, indicating that my model reduced prediction errors and was actually incorporating real information from my features.

After validation, I applied it to the testing data and added those predictions to the submissions file. I verified that the columns matched and remained in the correct order by *shot_id*.

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Conclusion

Overall, the model showed that shot type, defensive pressure, distance, shooter movement and game context all contributed to the predicted outcome. The visualizations helped me understand the model past the coefficient values, emphasizing the relationship between shot probability and distance. It proved that individual predictors cannot always provide accurate predictions when considered in isolation, because multiple factors rely on one another. So, having a measure of the interaction between certain predictors could really enhance predictive accuracy. This model performed better than the baseline on my validation split, but there are opportunities for improvement, including incorporating player-specific performance data, defender close-out speed/pressure, and feature-interaction measurements. With these additional features and further model refinement, the analysis could improve, providing a more complete representation of the factors affecting shot outcomes.